Shrinivasan Thevar

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Summary

AI Engineer specializing in **autonomous agent systems** and **loop engineering**: building **LLM-as-a-Judge** verification, closed-loop **eval loops**, and self-correcting rework cycles that keep **cloud agents** honest in production. Currently engineering **FangCheck**, the independent verification layer of a distributed AI workforce platform. Hands-on across **agentic RAG**, **LangGraph** multi-agent orchestration, and **vision-language** pipelines; previously shipped semantic search over **3,000+ videos** with **sub-2s latency** and **+10% retrieval accuracy**.

Technical Skills

Languages: Python, SQL, JavaScript, Bash/Linux

LLMs & GenAI: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, Tool Calling, Prompt Engineering, LLM-as-a-Judge, Eval Loops, Loop Engineering, Autonomous & Cloud Agents, LLM Autorouting, Rubric-Based Evaluation, Vision Language Models (VLM), Fine-Tuning (Unsloth, Hugging Face Transformers), Model Context Protocol (MCP), Whisper, OCR

Models & APIs: OpenAI API (GPT-5, GPT-4o-mini, text-embedding-3), Anthropic API, Groq (Llama 4 Scout, Llama 3.3 70B), TypeSafe AI Jev (System One decision model), BGE Embeddings

Frameworks: LangChain, LangGraph, FastAPI, Streamlit, React, Tailwind CSS, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV

Retrieval & Data: Milvus, FAISS, pgvector, FastEmbed, Cross-Encoder Re-ranking, Semantic Search, Docling, PostgreSQL, SQLAlchemy (async), MinIO

Platform & MLOps: Docker, Docker Compose, Git, GitHub, REST APIs, Idempotent Workflows, pytest, AWS, Tableau

Experience

Equations Work July 2026 – Present

AI Engineer – OpenWolfPack, FangCheck (Independent Verification)

- Worked on **OpenWolfPack**, a distributed **AI workforce platform** for founders and small businesses: an orchestrator agent takes a business goal over WhatsApp, decomposes it into tasks, and dispatches them to **cloud agents** and human specialists equipped with modular skill Kits.
- Engineered **FangCheck**, the platform's **autonomous LLM-as-a-Judge** verification service, through which **no output ships and no payment settles** without a passing verdict. It is skeptical by default ("wrong until proven right"): it grades every task against the contract, a versioned **rubric**, and the original instruction, and emits binary **PASS/FAIL verdicts** with criterion-linked findings. **Producer–judge independence** is enforced by separate models, credentials, and memory.
- Applied **loop engineering** to the rework cycle: FAIL findings re-enter the agent's warm-up context so retries converge instead of repeating, with a hard cap of **2 rework attempts** before a **human-in-the-loop** dispute; delivery and payment are gated idempotently on an **append-only verdict store**.
- Closed the **eval loop**: aggregated verdicts into first-pass, rework, and dispute rates per Kit and agent, feeding **agent reliability scores** into orchestrator matching and rubric tuning, with full vs. sampled review tiers and **cron-scheduled verification loops**.
- Integrated **Jev**, TypeSafe AI's **System One decision model**, into the platform's **AI gateway** for **autorouting**, making fast, structured routing decisions that send each agent request to the best-fit model.

Analyttica Datalabs Oct 2025 – Present

Data Science Intern (Generative AI) *Bangalore, India*

- Built an AI-driven video recommendation system using **semantic search** to retrieve relevant videos from natural-language queries across **3,000+ videos**.
- Engineered a **multimodal ingestion pipeline** combining **Vision Language Models** for frame captioning, **OCR** for on-screen text, and **Whisper** for audio transcription, turning raw video into searchable text.
- Designed a two-stage retrieval architecture (**FAISS** vector search + **Cross-Encoder** re-ranking), achieving **sub-2-second search latency** and a **10% increase in retrieval accuracy**.
- Architected a **RAG-based summarization** system in **LangChain** that generates context-aware, query-specific summaries for all **3,000+ indexed videos**.

Projects

Drone Security Analyst Agent – Autonomous Surveillance Monitoring *GitHub*

- Built an autonomous security-analyst agent that monitors a simulated drone feed (**49+** UCF-Crime CCTV clips with synthetic GPS, altitude, and waypoint telemetry) and raises real-time alerts.
- Designed a **LangGraph** pipeline (ingest, VLM analysis, history lookup, reasoning, alert/log, persist) with conditional routing so only alert-worthy or notable frames are logged.
- Used **Llama 4 Scout** (VLM) to extract objects, actions, clothing, and scene context per frame, and **Llama 3.3 70B** with deterministic rules to flag after-hours activity, loitering, repeat vehicles, and restricted-zone entry.
- Stored frames, events, and embeddings in **PostgreSQL + pgvector** with frames in **MinIO**, enabling semantic frame search and a **RAG chatbot** that calls the VLM as a tool on demand.
- Shipped a **FastAPI** REST API and **React** operator dashboard (live telemetry, alert banner, event log, session summaries), containerized with **Docker Compose** and covered by **pytest** plus 9 QA scenarios.

GroundTruth AI (Professor X) – Agentic Research Assistant *GitHub*

- Built an **agentic RAG** backend for scientific research that processes **1,000+ arXiv papers**, parsing PDFs into text chunks, tables, and figures with **Docling** and embedding them with **FastEmbed** (BGE) into **Milvus**.
- Designed a **6-agent LangGraph** system (Retriever, Recommendation, Repo, Buddy, Review, Writer) for citation-grounded answers, code-repository discovery, and scientific review writing, with on-demand **@agent** routing.
- Implemented **session-isolated** vector collections that merge into topic-level global stores, plus a self-healing **refetch loop** that ingests new papers when retrieval scores drop below **0.65**.
- Cut cost by routing pipeline agents to **GPT-5-nano** and synthesis to **GPT-5-mini**; added LLM topic classification and ID/DOI/title-hash deduplication, reducing manual research effort by **70%+**.

AutoML Pipeline Selector – No-Code Model Training App *GitHub*

- Built a **Streamlit** app that lets non-technical users upload a CSV/XLSX dataset, choose features, target, and task type, and train models without writing code.
- Automated preprocessing (missing-value imputation, label encoding, standard scaling, 80/20 split) and benchmarked **6 scikit-learn models** across classification and regression, ranked by **Accuracy/F1** or **R^2/MSE**.
- Exported the chosen model with its scaler and encoders as a reusable **joblib** bundle with ready-to-run inference code.

Education

Savitribai Phule Pune University July 2022 – June 2026
Bachelor of Engineering in Computer Science; CGPA: 8.22/10.0 *Pune, India*

Certifications

- Fundamentals of Machine Learning and Artificial Intelligence – AWS
- Artificial Intelligence Fundamentals – IBM
- Enterprise Data Science in Practice – IBM
- Getting Started with Data – IBM
- Deloitte Virtual Internship – Deloitte